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Individual Contribution Details for Milestone 1

The project work for Milestone 1 was divided among the three team members to ensure that the User Authentication & Role-Based Access Control and Project Management System modules were developed systematically. Each member contributed to specific functional areas while maintaining integration between authentication, user roles, project creation, scheduling, assignments, progress tracking, and project lifecycle management.

Team Member: Logeshwaran V

Modules Contributed

- User Authentication & Role-Based Access Control
- User Registration & Login
- User Account Management

Features Implemented

- User Registration
- User Login
- Email and Password Validation
- Secure Password Handling
- Password Hashing
- Confirm Password Validation
- Remember Me Functionality
- Forgot Password Functionality
- JWT Authentication

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- User Profile Information
- User Role Assignment
- Employee ID Management
- User Activation and Deactivation
- User Account Management
- Authentication Error Handling

Work Completed

Implemented the User Authentication and User Account Management functionality of the application. The work included developing the registration and login workflow, implementing secure password handling and validation, password hashing, confirm password validation, and generating JWT tokens after successful authentication.

The user account functionality was implemented to manage user profile information, employee IDs, roles, account activation and deactivation, and authentication-related errors. The authentication functionality provides the foundation for secure access to the remaining modules of the BuildTrack application.

Team Member: Dhanshri Chavhan

Modules Contributed

- Role-Based Access Control
- Project Creation & Project Planning

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- Project Scheduling

Features Implemented

Role-Based Access Control

- Role-Based Access Control (RBAC)
- Administrator Permissions
- Project Manager Permissions
- Site Engineer Permissions
- Contractor Permissions
- Worker Permissions
- Client Read-Only Access
- Role and Permission Validation

Project Creation & Project Planning

- Project Creation
- Project Name
- Project Code
- Project Category
- Client Details
- Project Description
- Project Location
- Estimated Budget
- Start Date
- Expected Completion Date

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- Project Priority
- Initial Project Status

Project Scheduling

- Activity Sequence
- Phase Duration
- Overall Project Schedule
- Planned Timeline

Work Completed

Implemented the Role-Based Access Control and Project Planning functionality. The work included defining permissions for the six application roles and ensuring that users can access only the modules and operations authorized for their respective roles.

The project planning functionality included project creation, project details, project categorization, client information, project description, location, estimated budget, project dates, priority, and initial project status.

Project scheduling functionality was also implemented to define construction phases, activity sequences, durations, and planned project timelines.

Team Member: Kattamuri N V S Sri Priya Saalini

Modules Contributed

- Project Assignment & Project Monitoring
- Project Milestone Management

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- Project Status, Update & Closure

Features Implemented

Project Assignment & Monitoring

- Project Manager Assignment
- Site Engineer Assignment
- Contractor Assignment
- Client Association
- Worker → Contractor → Project Relationship

Project Milestone Management

- Project Milestone Creation
- Milestone Planned Completion Date
- Milestone Actual Completion Date
- Planned vs. Actual Progress
- Milestone Completion Tracking

Project Status, Update & Closure

- Project Status Management
- Planning Status
- In Progress Status
- On Hold Status
- Completed Status
- Closed Status
- Project Information Update

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- Project Modification History
- Update Performed By
- Update Date and Time
- Project Closure Validation
- Project Completion Verification
- Final Inspection Verification
- Client Approval
- Financial Settlement Completion

Work Completed

Implemented the Project Assignment, Milestone Management, Project Status, Project Update, and Project Closure functionalities.

The work included assigning Project Managers, Site Engineers, Contractors, and Clients to projects, tracking project milestones using planned and actual completion dates, and maintaining project status throughout its lifecycle.

Project update history was maintained to provide transparency and accountability. Project closure functionality was implemented with appropriate validation for milestone completion, final inspection, client approval, financial settlement, project completion verification, and project handover.

Team Contribution & Integration

The three team members worked together to implement the User Authentication, Role-Based Access Control, and Project Management

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functionality required for Milestone 1.

User Authentication & User Management – Logeshwaran V

Logeshwaran implemented user registration, login, password security, JWT authentication, user profile management, and user account management. These functions provide the authentication foundation for the complete application.

RBAC & Project Planning – Dhanshri Chavhan

Dhanshri implemented role-based permissions, project creation, project information, project planning, and project scheduling. These functions ensure that projects are created with proper planning and that users can perform only authorized operations.

Project Assignment & Lifecycle Management – Saalini

Saalini implemented project assignments, milestone management, project status management, project updates, and project closure. These functions connect project planning with project execution, monitoring, completion, and finalization.

Together, these functionalities create the following connected workflow:

User Registration → Login & Authentication → Role Assignment →

Permission Validation → Project Creation → Project Planning → Project

Scheduling → Project Assignment → Milestone Tracking → Project Status

Updates → Project Completion → Project Closure

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Conclusion

Milestone 1 was completed through a clear and balanced division of responsibilities among the three team members.

Logeshwaran V focused on authentication and user account management, Dhanshri Chavhan focused on Role-Based Access Control, project creation, and project planning, while Kattamuri N V S Sri Priya Saalini focused on project assignments, milestones, project lifecycle management, updates, and closure.

The combined implementation establishes the foundation for the remaining modules of the BuildTrack application. Authentication and RBAC ensure secure and role-based access, while the Project Management System provides the project structure required by subsequent modules such as Site Progress Monitoring, Resource Management, Workforce Management, Procurement, Notifications, Reports, and Budget Management.

The completed Milestone 1 workflow is:

Authentication → RBAC → Project Creation → Project Planning →

Scheduling → Assignment → Milestones → Status Tracking → Project

Updates → Project Closure

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Individual Contribution Details for Milestone 2

The project work was divided among the team members to ensure that each member contributed to specific modules and features. The individual contributions are as follows.

Team Member: Logeshwaran V

Modules Contributed

- Site Activity Management
- Resource & Equipment Management

Features Implemented

- Site Activity Logs
- Machinery Maintenance
- Material Arrivals
- Safety Training
- Client Visits
- Government Inspections
- Quality Audits
- Accident Reports
- Contractor Meetings
- Equipment Servicing
- Resource and Category Management
- Add, Edit, and Delete Resources
- Equipment ID Management

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- Equipment Allocation to Projects
- Allocation and Return Dates
- Responsible Person Assignment
- Prevention of Duplicate/Overlapping Allocations
- Allocation History

Work Completed

Implemented the **Site Activity Management** and **Resource & Equipment Management** modules. The work included managing site activities, equipment and resources, project-wise equipment allocation, responsible personnel assignment, allocation and return dates, allocation history, and prevention of duplicate or overlapping equipment allocations.

Team Member: Dhanshri Chavhan

Modules Contributed

- Progress, Milestone & Delay Management
- Maintenance Scheduling

Features Implemented

- Weekly Progress Reports
- Work Completion Status
- Overall Project Completion Percentage
- Weekly Summary
- Progress Dashboard

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- Milestone Tracking
- Planned vs. Actual Progress
- Automatic Milestone Completion
- Completed and Pending Milestones
- Delay Records
- Delay Reasons
- Delay Duration
- Affected Work Category
- Timeline Impact
- Delay History
- Maintenance Records
- Last and Next Maintenance Dates
- Maintenance Type
- Service Engineer
- Maintenance Cost and Status
- Maintenance History
- Upcoming and Overdue Maintenance
- Maintenance Notifications

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Work Completed

Implemented the **Progress, Milestone & Delay Management** and **Maintenance Scheduling** modules. The work included monitoring project progress, tracking milestones, recording delays and their impact, and managing equipment maintenance schedules, maintenance history, costs, upcoming and overdue maintenance, and maintenance notifications.

Team Member: Kattamuri N V S Sri Priya Saalini

Modules Contributed

- Daily Progress Management
- Machinery Tracking & Resource Utilization

Features Implemented

- Daily Progress Reports
- Progress Photos
- Activity and Work Category
- Percentage of Work Completed
- Workers, Contractors, Machinery & Materials
- Weather, Safety & Quality Remarks
- Comments
- Equipment Location Tracking
- Current Project Tracking
- Equipment Status

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- Operating/Idle Status
- Resource Availability
- Operating Hours
- Idle Hours
- Utilization Percentage
- Utilization Reports

Work Completed

Implemented the **Daily Progress Management** and **Machinery Tracking & Resource Utilization** modules. The work included recording daily site progress, progress photos, work details, site conditions, and comments, along with tracking equipment location, current project, availability, operating hours, idle hours, equipment status, and utilization percentage.

Team Contribution & Integration

The three team members worked together to implement the project and site-progress-related functionality.

Site Activity & Resource Management:

Logeshwaran managed site activities and resource/equipment master data → equipment was made available for project allocation and tracking.

Progress, Milestone & Maintenance Management:

Dhanshri managed weekly progress, milestones, delays, and maintenance schedules → these functions supported project monitoring and equipment

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maintenance.

Daily Progress & Resource Utilization:

Saaliini managed daily progress and machinery utilization → daily site information and equipment usage data supported overall project monitoring.

Conclusion

The milestone was completed through a clear division of responsibilities among the three team members. Each member developed specific functional areas while maintaining integration between daily progress, site activities, project progress, equipment management, maintenance, and resource utilization.

Together, the team implemented the following connected workflow:

Daily Site Progress → Site Activities → Equipment & Resource

Management → Project Allocation → Maintenance → Resource Utilization

→ Progress & Monitoring

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MILESTONE EVALUATION – INDIVIDUAL CONTRIBUTIONS

Team Member	Assigned Responsibilities
Logeshwaran V	Workforce Management – Part 1; Procurement Management – Part 2
Dhanshri Chavhan	Workforce Management – Part 2; Procurement Management – Part 1
Kattamuri N V S Sri Priya	Workforce Management – Part 3; Procurement Management – Part 3
Saaliini	

MILESTONE 3 – WORKFORCE MANAGEMENT

The **Workforce Management** module manages workers and contractors, workforce allocation, shift scheduling, attendance, payroll, access control, and reporting.

Overall Workflow

Worker/Contractor Registration → Categorization → Assignment → Allocation → Shift Scheduling → Attendance → Working Hours → Payroll → RBAC → Reports

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Part 1 – Worker & Contractor Management

Assigned to: Logeshwaran V

Developed the worker and contractor master-management functions.

- **Worker Registration:** Worker ID, name, contact details, category, skill/work type, and status.
- **Workforce Categories:** Organizes workers based on skills such as mason, electrician, plumber, carpenter, painter, welder, and general worker.
- **Contractor Management:** Maintains contractor information and contractor–worker relationships.
- **Bulk Worker Registration:** Enables multiple workers to be added efficiently.
- **Worker Search and Filter:** Allows workers to be searched and filtered by worker details, category, contractor, and status.
- **Worker Status Management:** Identifies workers as active, inactive, or available for allocation.

Workflow

**Worker Registration → Category/Skill → Contractor Association → Status
→ Search/Filter → Available for Allocation**

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Part 2 – Workforce Allocation, Shift & Attendance

Assigned to: Dhanshri Chavhan

Developed workforce execution, shift scheduling, and attendance functions.

- **Project and Contractor Assignment:** Assigns workers based on skill, category, availability, and project requirements.
- **Workforce Allocation:** Allocates workers to appropriate projects and activities.
- **Assignment History:** Tracks worker, project, and contractor assignment history.
- **Shift Scheduling:** Manages shift name, start/end time, date, project, location, status, and assigned workers.
- **Worker–Project–Shift Relationship:** Connects workers with projects and scheduled shifts.
- **Attendance Tracking:** Records Present, Absent, and Leave status.
- **Check-in/Check-out:** Records worker attendance and calculates working hours.
- **Attendance Dashboard:** Displays total workers, present, absent, leave, working hours, shifts, and project-wise attendance.

Workflow

Project/Contractor Assignment → Allocation → Shift → Worker

Assignment → Check-in → Work → Check-out → Working Hours →

Attendance Dashboard

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Part 3 – Payroll, RBAC & Module Integration

Assigned to: Kattamuri N V S Sri Priya Saalini

Developed payroll monitoring, access control, integration, and reporting functions.

- **Payroll Monitoring:** Uses pay rate, working days, working hours, overtime, and leave information.
- **Estimated Pay:** Calculates estimated payroll based on workforce attendance and working information.
- **Payroll Status:** Supports statuses such as Pending, Processing, and Paid.
- **Role-Based Access Control (RBAC):** Provides access based on roles such as Admin, Project Manager, Site Engineer, and other authorized users.
- **Project-based and Contractor-based Access:** Controls workforce operations based on project and contractor permissions.
- **Module Integration:** Integrated Workforce Management with Modules 1–5 and Module 7.
- **Workforce Reports and Dashboard:** Provides information about workforce, contractors, allocation, shifts, attendance, working hours, and payroll.

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Workflow

Attendance Data → Working Hours/Days → Overtime & Pay Rate →

Estimated Payroll → Payroll Status → Reports

RBAC applies across all authorized workforce operations.

PROCUREMENT MANAGEMENT

The **Procurement Management** module manages vendors, procurement requests, purchase orders, suppliers, invoices, inventory integration, access control, and reporting.

Overall Workflow

Vendor Registration → Procurement Request → Approval → Vendor

Selection → Purchase Order → Delivery → Invoice → Payment →

Inventory Update → Reports

Part 1 – Vendor & Procurement Request Management

Assigned to: Dhanshri Chavhan

Developed vendor/supplier management and procurement request functions.

- **Vendor/Supplier Registration:** Maintains vendor details, contact information, and supplied products/services.
- **Vendor Categories:** Organizes vendors such as cement, steel, electrical, plumbing, equipment, and safety suppliers.
- **Procurement Categories:** Manages different procurement categories.

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- **Procurement Request Creation:** Creates requests with items, quantity, required date, priority, and project association.
- **Request Status Management:** Tracks procurement request status.
- **Approval Workflow:** Supports Draft → Submitted → Pending Approval → Approved/Rejected → Purchase Order.

Workflow

Vendor Registration → Category/Product Details → Material Requirement → Request → Items/Quantity → Date/Priority → Project → Submit →

Approval

Part 2 – Purchase Orders & Supplier Management

Assigned to: Logeshwaran V

Developed purchase order creation and supplier/order tracking functions.

- **Vendor Selection:** Selects the appropriate vendor after procurement request approval.
- **Purchase Order Creation:** Creates POs with vendor, project, PO details, items, quantities, prices, taxes, total amount, delivery date, and status.
- **PO Item Management:** Manages item quantities, unit prices, taxes, and total amount calculations.
- **Expected Delivery Date:** Tracks expected delivery dates for purchase orders.
- **Purchase Order Status:** Tracks the current status of purchase orders.

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- **Supplier–Purchase Order Relationship:** Maintains the relationship between suppliers and purchase orders.
- **Procurement Request–Purchase Order Relationship:** Connects approved procurement requests with purchase orders.
- **Project–Purchase Order Relationship:** Associates purchase orders with the relevant projects.
- **Order Tracking:** Tracks ordered items, suppliers, quantities, delivery information, and order progress.

Workflow

Approved Request → Vendor Selection → Purchase Order →

Items/Quantity → Price/Tax → Delivery Date → PO Issue → Supplier

Processing → Delivery Tracking

Part 3 – Invoice, Inventory Integration & Access Control

Assigned to: Shalini

Developed invoice tracking, inventory integration, access control, and procurement reporting.

- **Invoice Tracking:** Maintains invoice number, invoice date, amount, due date, payment status, and invoice status.
- **Purchase Order–Invoice Relationship:** Links invoices with their corresponding purchase orders.
- **Inventory Integration:** Connects procurement with material inventory.

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- **Procurement Workflow Integration:** Material Requirement → Inventory Check → Procurement Request → Purchase Order → Material Received → Inventory Update.
- **Role-Based Access Control:** Provides procurement access based on authorized roles.
- **Project-based Procurement Access:** Restricts procurement operations according to project permissions.
- **Approval Permissions:** Controls procurement approval operations.
- **Module Integration:** Integrated Procurement Management with Modules 1, 2, 5, 8, 9, and 11.
- **Procurement Dashboard and Reports:** Provides information about vendors, requests, POs, deliveries, invoices, payments, and project-wise procurement.

Workflow

**Purchase Order → Material Delivery → Invoice → Verification →
Payment Tracking → Inventory Update → Procurement Reports**

INDIVIDUAL CONTRIBUTION SUMMARY

Logeshwaran V

Workforce Management – Part 1

- Worker Registration
- Workforce Categories

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- Contractor Management
- Contractor–Worker Relationship
- Bulk Worker Registration
- Worker Search and Filter
- Worker Status Management

Procurement Management – Part 2

- Vendor Selection
- Purchase Order Creation
- PO Items and Quantities
- Pricing and Taxes
- Delivery Date
- Purchase Order Status
- Supplier–PO Relationship
- Procurement Request–PO Relationship
- Project–PO Relationship
- Purchase Order Tracking

Overall Contribution

Focused on **workforce master data management** and **procurement purchase-order and supplier operations**.

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Dhanshri Chavhan

Workforce Management – Part 2

- Project and Contractor Assignment
- Workforce Allocation
- Assignment History
- Shift Scheduling
- Worker–Project–Shift Relationship
- Attendance Management
- Check-in/Check-out
- Working Hours Calculation
- Attendance Dashboard

Procurement Management – Part 1

- Vendor/Supplier Management
- Vendor Registration
- Vendor Categories
- Products/Services
- Procurement Categories
- Procurement Requests
- Items and Quantity
- Required Date and Priority
- Project Association
- Request Status

Documentation

- Approval Workflow

Overall Contribution

Focused on **workforce execution and attendance management** and **vendor/procurement request management**.

Shalini

Workforce Management – Part 3

- Payroll Management
- Pay Rate
- Working Days
- Working Hours
- Overtime
- Leave
- Estimated Pay
- Payroll Status
- Role-Based Access Control
- Project/Contractor Access
- Permissions
- Module Integration
- Workforce Reports and Dashboard

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Procurement Management – Part 3

- Invoice Tracking
- Invoice Details
- Due Date
- Payment Status
- Invoice Status
- PO–Invoice Relationship
- Inventory Integration
- Role-Based Access Control
- Project-based Access
- Approval Permissions
- Module Integration
- Procurement Reports and Dashboard

Overall Contribution

Focused on **payroll, security, invoice management, inventory integration, reporting, and module integration.**

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TEAM CONTRIBUTION & INTEGRATION

Workforce Management

Logeshwaran V provides worker and contractor master data → **Dhanshri Chavhan** uses the data for workforce allocation, shifts, and attendance → **Shalini** uses attendance and working-hour data for payroll, permissions, and reporting.

Workforce Flow

Worker/Contractor Management → **Allocation** → **Shift Scheduling** → **Attendance** → **Working Hours** → **Payroll** → **RBAC** → **Reports**

Procurement Management

Dhanshri Chavhan manages vendors and procurement requests → **Logeshwaran V** converts approved requirements into purchase orders and manages supplier/order tracking → **Shalini** manages invoices, inventory integration, access control, and reporting.

Procurement Flow

Vendor Management → **Procurement Request** → **Approval** → **Purchase Order** → **Delivery** → **Invoice** → **Inventory Integration** → **RBAC** → **Reports**

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CONCLUSION

The milestone was completed through a clear division of responsibilities among the three team members. Each member developed a specific functional area while maintaining integration with the other parts of the system.

Together, the team implemented two connected workflows:

Workforce Management

Worker/Contractor Management → Workforce Allocation → Shift & Attendance → Payroll → RBAC → Reports

Procurement Management

Vendor Management → Procurement Request → Purchase Order → Invoice → Inventory Integration → RBAC → Reports

The successful integration of these components provides a complete and connected workflow for **workforce and procurement management within the BuildTrack construction management system.**

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Individual Contribution Details for Milestone 4

The project work for Milestone 4 was divided among the team members, with each member responsible for the development and implementation of a specific module.

Module 8 – Notification System

Team Member: Dhanshri Chavhan

Individual Contribution

Dhanshri Chavhan was responsible for the development and implementation of

Module 8 – Notification System.

Features Added

- Project-related Notifications
- Task Notifications
- Procurement Notifications
- Attendance Notifications
- Milestone Notifications
- Read and Unread Notification Status
- Notification Bell and Dropdown
- User-based Notification Access
- JWT-based Authorization

Work Completed

- Developed the notification system for different project activities.
- Implemented notification-related backend functionality.

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- Integrated notifications with the PostgreSQL database.
 - Added notification display functionality in the Angular frontend.
 - Implemented read and unread notification status.
 - Integrated notifications with project, task, procurement, attendance, and milestone activities.
 - Implemented user-based notification access and authorization.
 - Tested notification creation, display, and read/unread functionality.
-

Module 9 – Dashboard & Analytics

Team Member: Logeshwaran V

Individual Contribution

Logeshwaran V was responsible for the development and implementation of

Module 9 – Dashboard & Analytics.

Features Added

- Admin Dashboard
- Project Manager Dashboard
- Project Progress Analytics
- Milestone Analytics
- Workforce Analytics
- Resource Utilization Analytics
- Procurement Analytics
- Budget and Cost Analytics

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- Overall Project Completion Percentage
- Planned vs. Actual Progress
- Live PostgreSQL Data Integration
- Role-based Dashboard Access

Work Completed

- Developed dashboard functionality for different user roles.
 - Integrated live data from the PostgreSQL database.
 - Implemented project progress and completion calculations.
 - Added milestone, workforce, resource, procurement, and budget metrics.
 - Implemented role-based dashboard access.
 - Connected backend dashboard APIs with the Angular frontend.
 - Implemented project-assignment-based data access for Project Managers.
 - Verified dashboard metrics using live project data.
-

Module 10 – Reports & Documentation

Team Member: Abhay

Individual Contribution

Abhay was responsible for the development and implementation of **Module 10 – Reports & Documentation**.

Features Added

- Project Progress Report
- Resource Utilization Report

Documentation

- Workforce Report
- Procurement Report
- Budget & Cost Report
- PDF Report Generation
- Excel Report Generation
- JSON Report Generation
- Project-based Report Filtering
- Role-based Report Access

Work Completed

- Developed reporting functionality using live project data.
 - Implemented report generation APIs.
 - Added PDF report generation using ReportLab.
 - Added Excel report generation using openpyxl.
 - Added JSON report generation.
 - Integrated report data with the relevant project modules.
 - Implemented project-based filtering.
 - Implemented role-based access control for reports.
 - Verified that unauthorized users cannot access restricted project reports.
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Module 11 – Budget & Cost Management

Team Member: Kattamuri N V S Sri Priya Saalini

Individual Contribution

Saalini was responsible for the development and implementation of **Module 11 – Budget & Cost Management**.

Features Added

- Project Budget Management
- Budget Allocation
- Cost Tracking
- Procurement Cost Tracking
- Purchase Order Cost Integration
- Budget Utilization
- Remaining Budget Calculation
- Planned vs. Actual Cost Analysis
- Cost Percentage Analysis
- Project-wise Cost Monitoring
- Role-based Access Control

Work Completed

- Developed project budget management functionality.
- Implemented budget allocation and cost tracking.
- Integrated procurement and purchase order costs with budget management.

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- Implemented budget utilization and remaining budget calculations.
 - Added planned vs. actual cost analysis.
 - Implemented project-wise cost monitoring.
 - Integrated budget and cost information with the dashboard.
 - Implemented role-based access control.
 - Tested and verified budget and cost management functionality.
-

Team Contribution & Integration

The team members contributed to different modules while maintaining integration across the BuildTrack system.

Module 8 – Notification System:

Dhanshri developed notifications for project, task, procurement, attendance, and milestone activities and integrated them with user authorization.

Module 9 – Dashboard & Analytics:

Logeshwaran developed role-based dashboards and integrated live PostgreSQL data for project progress, milestones, workforce, resources, procurement, and budget analytics.

Module 10 – Reports & Documentation:

Abhay developed project reports in PDF, Excel, and JSON formats with project-based filtering and role-based access control.

Module 11 – Budget & Cost Management:

Saaliini developed budget allocation, cost tracking, procurement cost integration,

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budget utilization, planned vs. actual cost analysis, and project-wise cost monitoring.

Conclusion

Milestone 4 was completed through a clear division of responsibilities among the four team members. Each member contributed to a specific module while maintaining integration with the overall BuildTrack system.

Together, the team implemented the following connected workflow:

Project Activities → Notifications → Dashboard & Analytics → Reports & Documentation → Budget & Cost Management

The modules work together using live project data, role-based access control, PostgreSQL integration, and cross-module functionality to provide a complete construction project management solution.